STAT 505 Assignment Solutions [L1 & L2]

1. The following $n \times p$ data matrix contains $n = 3$ observations for each of $p = 2$ variables X_1 (column 1) and X_2 (column 2).

$$\begin{bmatrix} 3 & 18 \\ 8 & 21 \\ 4 & 20 \end{bmatrix}$$

Calculate each of the following. You can check your work with software, but you should show by hand how the numbers in this case fit into the formula for each.

(a) The sample mean vector $\bar{\mathbf{x}}$ *Hint: this is a* 2×1 *vector consisting of the sample means for both variables*

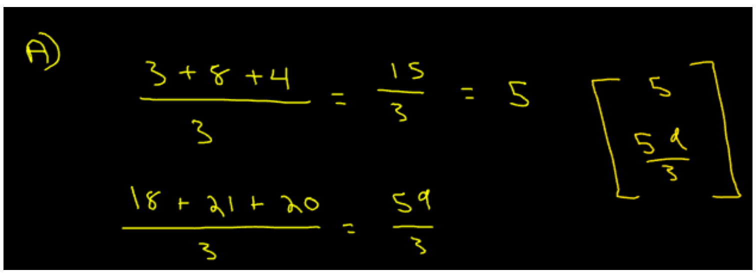

Figure 1: Handwritten solution for question 1a.

As shown in Figure 1, the sample mean vector is:

$$\bar{x} = \begin{bmatrix} 5 \\ \frac{59}{3} \end{bmatrix}$$

(b) The sample covariance matrix $\mathbf{S}$. Note that we divide by $n-1$ for this (not n). *Hint: this is a* 2×2 *matrix consisting of the sample variances for both variables on the diagonals and their covariance on the off-diagonals*

As shown in Figure 2, the sample sample covariance matrix $\mathbf{S}$ is:

$$S = \begin{bmatrix} 7 & 3.5 \\ 3.5 & 2..333 \end{bmatrix}$$

(c) The sample correlation matrix $\mathbf{R}$

As shown in Figure 3, the sample sample correlation matrix $\mathbf{R}$ is:

$$R = \begin{bmatrix} 1 & 0.866 \\ 0.866 & 1 \end{bmatrix}$$

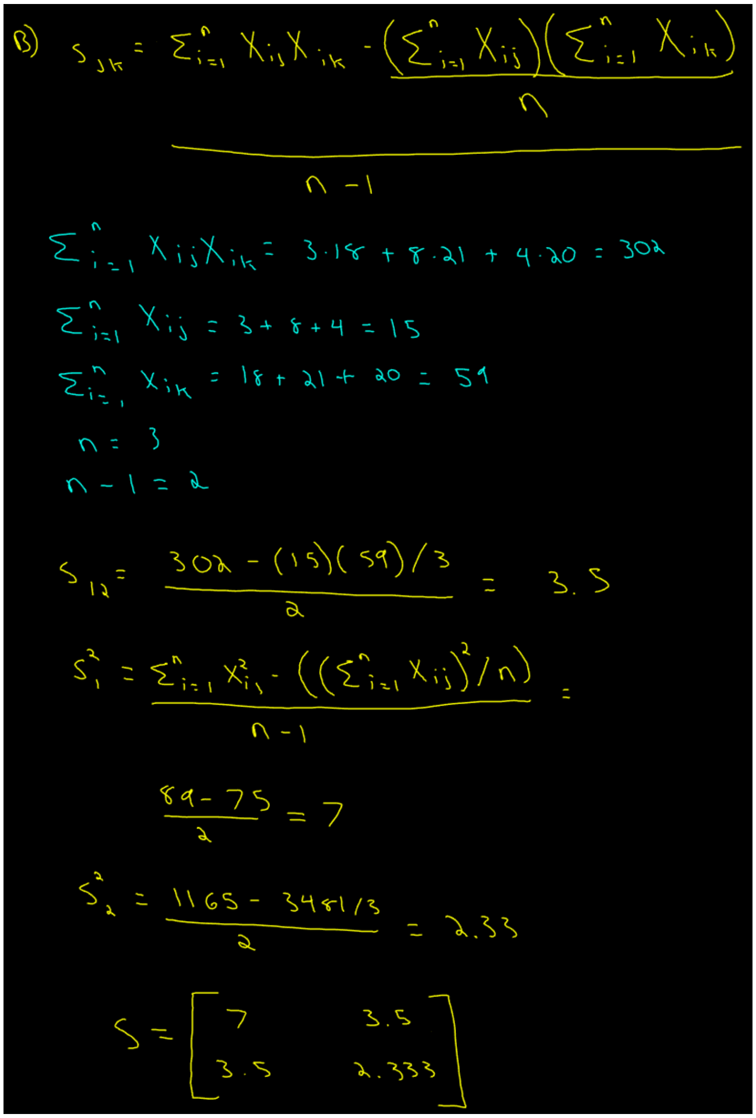

Figure 2: Handwritten solution for question 1b.

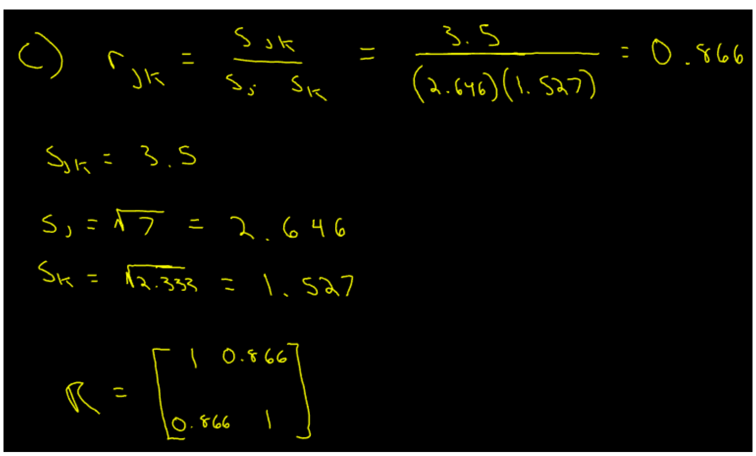

Figure 3: Handwritten solution for question 1c.

2. Let $\mathbf{X}' = [X_1, X_2]$ denote the vector of variables from above, and define $\mathbf{Y}' = [Y_1, Y_2]$ by

$$Y_1 = (X_1 + X_2)/2$$

and

$$Y_2 = X_2 - 2X_1$$

(a) Find the coefficient matrix $\mathbf{A}$ such that $\mathbf{Y} = \mathbf{AX}$.

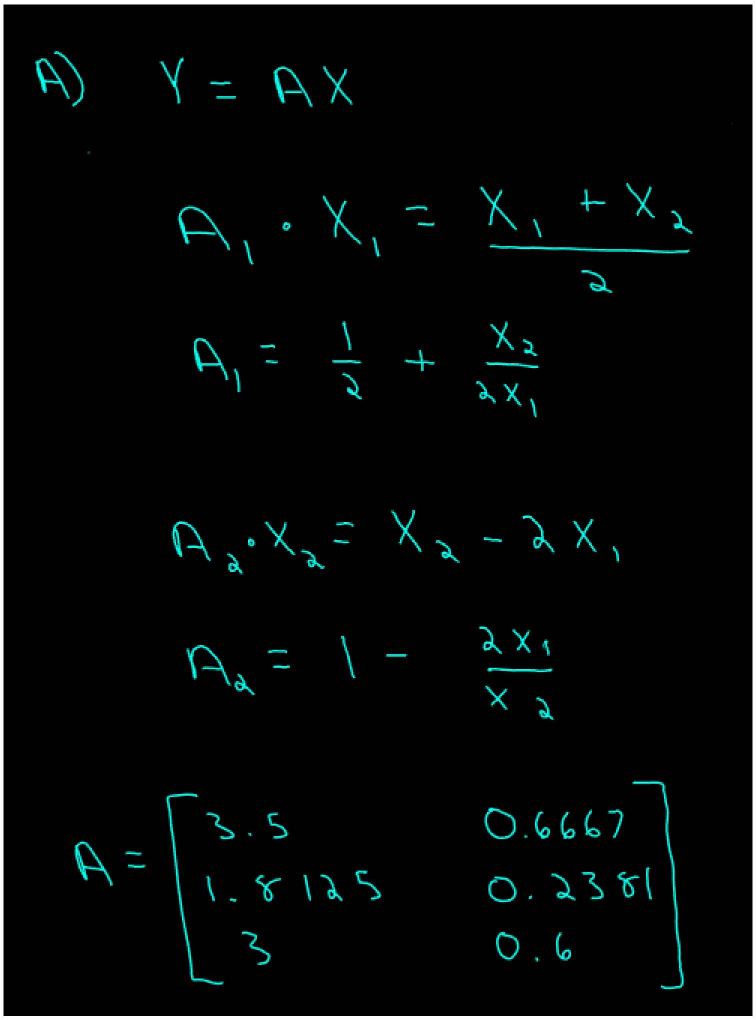

Figure 4: Handwritten solution for question 2a.

As shown in Figure 4, the coefficient matrix $\mathbf{A}$ is:

$$A = \begin{bmatrix} 3.5 & 0.6667 \\ 1.8125 & 0.2381 \\ 3 & 0.6 \end{bmatrix}$$

(b) Find the sample mean vector and covariance matrix for $\mathbf{Y}$

B)

$$Y = \begin{bmatrix} 10.5 & 12 \\ 14.5 & 5 \\ 12 & 12 \end{bmatrix}$$

$$\bar{y} = \begin{bmatrix} 12.3 \\ 9.6667 \end{bmatrix}$$

$$S_{12} = \frac{342.5 - \frac{37 \cdot 29}{3}}{2} = -7.583$$

$$s_1^2 = \frac{464.5 - 1369/3}{2} = 4.0833$$

$$s_2^2 = \frac{313 - 841/3}{2} = 16.333$$

$$S_y = \begin{bmatrix} 4.0833 & -7.583 \\ -7.586 & 16.333 \end{bmatrix}$$

Figure 5: Handwritten solution for question 2b.

As shown in Figure 5, the sample mean vector and covariance matrix for $\mathbf{Y}$ is: [1]

$$\bar{y} = \begin{bmatrix} 12.3 \\ 9.666 \end{bmatrix}$$

$$S_y = \begin{bmatrix} 4.0833 & -7.583 \\ -7.583 & 16.333 \end{bmatrix}$$

3. The data "companies.csv" is available on our Canvas page and can be read into SAS with the commands below. The variables (columns) are X_1, X_2, and X_3.

```
data company;
infile "/companies.csv" delimiter="," firstobs=2;
input x1 x2 x3;
run;
```

Define new variables $Y_1 = X_1 + X_2 + X_3$ and $Y_2 = X_3 - X_1$. Find the mean and variance for each of Y_1 and Y_2. Also, find the correlation between Y_1 and Y_2.

This is the SAS code used in SAS Online Studio for this question:

```
data company;
infile "/home/u64433460/companies.csv" delimiter="," firstobs=2;
input x1 x2 x3;
y1 = x1 + x2 + x3;
y2 = x3 - x1;
run;

proc means data=company mean var;
    var y1 y2;
run;

proc corr data=company plots=none;
    var y1 y2;
run;
```

The SAS procedures resulted in the following tables that present the mean, variance, and the covariance in figures 6 and 7.

Variable	Mean	Variance
y1	881.2180000	171904.58
y2	555.3080000	315682.64

Figure 6: SAS solution for question 3.1, showing the mean and the variance for Y_1 and Y_2.

[1] after handwriting, I realized the transposing mistake and corrected in the latex document

Pearson Correlation Coefficients, N = 10 Prob > |r| under H0: Rho=0		
	y1	y2
y1	1.00000	0.97968 <.0001
y2	0.97968 <.0001	1.00000

Figure 7: SAS solution for question 3.2, showing the covariance between Y_1 and Y_2.